

Synthesis & Recommended Approach

RSNA Knee MRI Abnormality Detection — Kaggle Competition

Based on a review of 8 papers spanning knee MRI deep learning, medical vision-language pretraining, and radiology-report NLP, organized around four design questions: MRI handling, report/NLP processing, multimodal fusion, and training strategy. The final section proposes one combined, practical pipeline for the competition.

1. MRI Side — Sequence Selection, Dimensionality, and Plane Fusion

What the literature converges on

Two clearly different strategies emerge for handling multi-plane, multi-sequence knee MRI, and they trade off against each other rather than one simply beating the other.

Full 3D, single-sequence, segment-then-classify (Astuto et al., 2021). A two-stage V-Net pipeline first segments 11 anatomical regions of interest from a single 3D fast-spin-echo sequence, crops bounding-box subvolumes around each, then runs 17 separate 3D CNN classifiers — one per tissue/severity decision — organized as a hierarchical binary-then-severity cascade. This achieves strong AUCs (0.83–0.93) and demonstrably improves human inter-rater agreement, but it depends on a single, high-resolution 3D sequence and does not fuse across planes; the paper's own limitations section notes that collateral ligaments were missed because only a sagittal sequence was used.

Multi-plane, multi-sequence, cross-attention fusion (Qiu et al., 2024 — CoPAS). Rather than one 3D volume, this approach takes sagittal, coronal, and axial PDW volumes plus T1W/T2W contrast sequences, and — critically — compensates for thin-slice, anisotropic sequences by geometrically rotating one plane's volume to synthesize a plausible view of another plane, rather than discarding information or relying on interpolation. Each plane is encoded by a moderate-depth ResNet3D-18 (not a very deep 3D backbone), and cross-plane/cross-sequence attention integrates information before a final probability-matrix fusion step (see Section 3).

The commentary by Li & Chang (2021) frames the underlying reason both approaches matter: single-tissue, single-facet AI models are "individual puzzle pieces," and integrating multiple tissues, severity grading, and (implicitly) multiple planes into one coherent system is what has clinical value — which is exactly the shape of the RSNA task (12 findings, one study, multiple series).

Implication for RSNA

The RSNA dataset gives multiple MRI series per study but not necessarily the uniform three-plane, five-sequence structure CoPAS assumes, and, unlike Astuto et al.'s single-cohort 3D CUBE sequence, RSNA series will vary in slice count, spacing, and orientation across the 16 contributing sites. This argues against copying either paper's exact input assembly and instead building a lighter version of CoPAS's idea: a per-plane/per-sequence encoder (2.5D rather than full 3D, given code-competition runtime limits), with a lightweight cross-plane attention step used only where more than one relevant series is actually available for a study, falling back to single-plane classification otherwise. Astuto et al.'s segment-then-crop preprocessing is worth adapting at a coarse level — not full anatomical segmentation, but a similar principle of localizing the joint/region before classification to reduce background noise the model has to learn through.

2. NLP Side — Extracting Abnormalities from Reports, Negation, and Weak Labels

Entity/label extraction approaches, compared

Fixed ontology + existing NLP tool (Wang et al., 2022 — MedCLIP). Uses MetaMap against a UMLS-defined set of 14 finding types, and — importantly — maps structured image-level labels into the same UMLS concept space as extracted report entities, so images-with-labels and reports-with-free-text become comparable. This shared-vocabulary trick is the cleanest solution across all 8 papers to a problem RSNA has directly: some of your ~4,407 studies likely have different amounts of structured vs. free-text signal, and a shared concept space lets you combine them instead of treating them as two disconnected label sources.

Curated + ontology-expanded keyword list (Das et al., 2025). Starts from a small, manually curated term list (roughly 100 terms) and mechanically expands it via SNOMED-CT synonym relations (using PyMedTermino2) to ~210 terms, then matches LLM-extracted findings against this list with fuzzy string matching. This is a lighter-weight, more auditable alternative to MetaMap — worth using if you want direct control over your finding vocabulary rather than trusting a black-box entity tagger.

LLM-based weak-label bootstrapping (Das et al., 2025). A genuinely reusable two-phase recipe: Phase I uses a pretrained instruction-tuned LLM (Mistral-7B, in this paper's case) with few-shot prompting to generate finding labels from unlabeled reports; Phase II fine-tunes the same model class on its own Phase-I output via LoRA. The paper's own ablation is a useful, non-obvious finding: the general-domain Mistral outperformed the biomedical BioMistral on this task, so domain-specific pretraining should not be assumed to win without testing.

Negation and uncertainty — the clearest gap across the literature

This is the one place where every relevant paper in the set agrees, by omission or explicit admission, that the problem is unsolved:

- MedCLIP's own limitations section states its entity extraction "still encounters failure cases where incorrect semantic tags are detected or missing detection of negation or uncertainty phrases," and proposes noisy-label learning as an unimplemented remedy.
- Das et al. (2025) document this failure concretely with real examples: an instruction-tuned LLM extracted "bowel obstruction" from a report that stated "No evidence for intestinal obstruction," and failed to flag genuine findings buried in "No significant interval change" phrasing describing a known chronic fracture.
- Neither ConVIRT nor GLoRIA (Zhang et al., 2022; Huang et al., 2021) attempt negation handling at all — their text pipelines are sentence-sampling and word-piece tokenization only.

The practical conclusion: none of these 8 papers can be relied on, as-is, to solve negation for the RSNA reports. A dedicated rule-based negation pass (e.g., a NegEx/NegBio-style scope-based negation detector, cited in passing inside the MedCLIP and Das et al. reference lists but not implemented in either paper) needs to sit between entity extraction and label assignment — this is a genuine gap the RSNA project needs to fill itself, not something to borrow wholesale from any one source.

3. Multimodal Side — Combining Image and Report Representations

Three points on a spectrum, not a binary choice

Global-only contrastive fusion (Zhang et al., 2022 — ConVIRT). One embedding per image, one per report, aligned via a symmetric InfoNCE-style loss. Simple, and the paper's central empirical result — 10x label efficiency versus ImageNet initialization on downstream classification — is arguably the single most transferable finding for a data-scarce project like RSNA: pretraining on paired image-report data, even without dense per-finding labels, meaningfully reduces how many gold labels you need afterward.

Global + local cross-attention fusion (Huang et al., 2021 — GLoRIA). Adds a second loss that aligns attention-weighted image sub-regions against individual report words, without needing a pretrained object detector — a real middle path between early fusion (concatenate everything up front) and late fusion (combine only final predictions). At inference, GLoRIA simply averages global and local similarity scores — a low-risk fallback fusion rule if a more elaborate scheme underperforms.

Plane-aware probability-matrix fusion (Qiu et al., 2024 — CoPAS). A genuinely different late-fusion idea for combining more than two sources: rather than averaging independent predictions, it takes the outer product of the per-plane probability vectors and treats the diagonal as the base joint prediction, refined by a small learned correction term. This generalizes naturally beyond image planes to combining an image-only prediction with a report-only prediction.

Decoupled, soft-target contrastive fusion (Wang et al., 2022 — MedCLIP). Perhaps the most directly useful idea for RSNA's actual data shape: instead of requiring every training example to be a matched (image, report) pair, MedCLIP treats all labeled-only images and all report-only texts as a combinatorial pool, and replaces hard positive/negative contrastive labels with soft targets derived from entity/label similarity — directly solving the false-negative problem that arises when two different patients' reports describe the same finding but get treated as a mismatch.

Implication for RSNA

RSNA's data shape — a small gold-labeled subset, a much larger report-only or partially-labeled set, and images that must be classifiable without a report at inference time — makes MedCLIP's decoupled/soft-target pairing and ConVIRT's "pretrain without dense labels, fine-tune on what little gold data exists" strategy the two most load-bearing ideas here, with GLoRIA's local attention layer as an add-on once a working global pipeline exists, not a starting point.

4. Training Side — Imbalance, Noisy Labels, and Macro-AUC

Hierarchical decomposition to fight class imbalance (Astuto et al., 2021). Splitting a multi-class severity problem into a binary normal/abnormal gate followed by a conditional severity classifier reduces the effective imbalance each individual model has to learn, and is a large part of why their per-class sensitivities held up on minority severity classes.

Multi-task learning with shared features helps rare classes — but not uniformly (Qiu et al., 2024). CoPAS's single joint model across all 12 findings improved performance specifically on its rarest class (patellar retinaculum injury), attributed to shared feature representations, but the same paper honestly reports multi-task learning hurting 4 of 12 findings through negative transfer between visually similar classes (e.g., effusion vs. cyst). The combined loss used — per-branch binary cross-entropy plus Focal Loss on the fused final prediction — is a concrete, reusable recipe for simultaneously supervising intermediate branches and focusing the final head on hard/rare examples.

Soft, similarity-derived targets over hard contrastive labels for noisy pairing (Wang et al., 2022). Directly relevant to optimizing under label noise: cross-entropy against a softmax-normalized semantic-similarity target, rather than treating any non-identical pair as a hard negative, measurably reduces the harm from false negatives in a weakly-paired setting.

Weak-label bootstrapping and LLM-judge validation at scale (Das et al., 2025). Fine-tuning on your own model's Phase-I weak labels, then validating with a combination of lexical-overlap metrics and LLM-as-judge scoring, is a practical way to sanity-check label quality across thousands of report-only studies without hand-checking each one.

Distill-then-drop-a-modality at inference (Lu & Chen, 2022 — cross-domain analogy, used cautiously). Outside medical imaging entirely, this ASR paper's training-time-only knowledge distillation from a frozen language model into an acoustic encoder — discarding the language-model branch entirely at inference — is a structurally relevant pattern if RSNA's test set does not guarantee a report is available at inference: train an image encoder to match report-derived (BERT-style) targets during training, then run image-only at test time.

Macro-ROC-AUC specifically. None of these 8 papers optimize macro-AUC as their primary target (most report AUC per class descriptively, and Astuto et al. and Qiu et al. use BCE/Focal Loss rather than an AUC-surrogate

loss), so no direct macro-AUC optimization trick can be honestly attributed to this reading list. The practical bridge is: use per-class weighted BCE/Focal Loss during training (as CoPAS does) balanced by class prevalence, and treat macro-AUC purely as the evaluation metric guiding validation-set model selection and threshold/calibration choices, rather than as a directly differentiable training objective — this is the standard, defensible way to target a non-differentiable ranking metric with a differentiable loss, and is consistent with (though not verbatim from) how each paper's own evaluation protocol is separated from its training loss.

5. Recommended Approach for the RSNA Project

The recommendation below is a synthesis: no single piece is copied wholesale from any one paper, and several pieces (the fallback rule for missing planes, the negation pass, the modality-dropout inference design) are not fully specified in any of the 8 sources and represent a combination and extension of their ideas for this specific competition's data shape.

Stage 1 — Weak-label construction from reports (before any model training)

- Define a fixed 12-finding vocabulary matching the competition's target findings (following the fixed-vocabulary pattern common to MedCLIP and Das et al., rather than an open-ended entity tagger).
- Extract candidate finding mentions with a lightweight, auditable keyword/synonym list (Das et al.'s SNOMED-expansion approach), rather than a full MetaMap pipeline, for easier debugging on a solo project timeline.
- Run a dedicated negation/uncertainty pass (a NegEx/NegBio-style scope detector) over every extracted mention before it becomes a label — this step exists in none of the 8 papers as implemented, and is the single highest-value original addition given how consistently every paper that touched this problem either failed at it or admitted skipping it.
- Where both a structured label and a free-text mention exist for the same finding, reconcile them into one shared concept space (MedCLIP's UMLS-mapping idea) rather than keeping two disagreeing label sources.

Stage 2 — Image pipeline

- Localize the joint/region of interest coarsely before classification (adapted, lighter version of Astuto et al.'s segment-then-crop, without their full 11-region V-Net if compute/time is tight).
- Encode each available plane/sequence with a moderate 2.5D or ResNet3D-18-scale backbone (CoPAS's choice, not a very deep 3D network) to fit Kaggle's code-competition runtime limit.
- Apply cross-plane attention only where more than one relevant series exists for a study; fall back to single-plane classification otherwise, rather than assuming CoPAS's full five-volume input is always available.

Stage 3 — Multimodal training

- Pretrain the image encoder against report-derived soft targets using a MedCLIP-style semantic-similarity loss (not hard InfoNCE), using every image — gold-labeled or not — paired combinatorially against every available report or report-derived label vector, to make full use of the ~4,349 report-only studies rather than only the 58 gold-labeled ones.
- Add a GLORIA-style local attention term once the global pipeline is stable and only if time remains — it is a refinement, not a prerequisite.
- Because the competition's held-out test studies may not reliably include a usable report, structure this as training-time-only multimodal supervision: the report branch is a distillation target during pretraining (as in Lu & Chen's ASR analogy), and only the image branch is required at inference — this removes any risk of building a pipeline that silently depends on test-time report availability.
- Fine-tune the resulting image encoder on the small gold-labeled subset with a hierarchical binary-then-severity structure (Astuto et al.) for whichever findings have graded severity, and a joint multi-task head

with combined per-finding BCE + Focal Loss (CoPAS's loss design) across all 12 findings to share signal between correlated/rare findings.

Stage 4 — Validation and model selection

- Track macro-ROC-AUC as the selection metric on a held-out validation split stratified by finding prevalence, since none of the source papers' training losses directly optimize it — losses stay classification-based (BCE/Focal), AUC stays the evaluation lens.
- Use LLM-as-judge scoring (Das et al.'s G-Eval/Prometheus approach) as a cheap sanity check on weak-label quality across the large report-only pool before trusting it for pretraining.
- Given the efficiency track the 2026 competition specifically introduced, prefer the smaller backbones and fewer parallel branches suggested above (2.5D over full 3D, ResNet3D-18-scale, single joint multi-task head over 17 separate per-tissue models) as a deliberate trade-off, consistent with CoPAS's own finding that a moderate-depth shared encoder can match or beat deeper single-task ensembles.

6. On Originality and Attribution

Every recommendation above traces back to a specific paper's idea, cited by author and year, but none of the four stages is a direct copy of any single paper's pipeline: Astuto et al.'s cropping and hierarchical severity structure is combined with CoPAS's plane-fusion and loss design; MedCLIP's decoupled soft-target pretraining is combined with ConVIRT's data-efficiency framing and Lu & Chen's train-time-only distillation pattern (borrowed as a cross-domain analogy, not a medical-imaging precedent); the negation-handling stage is original to this synthesis because it is the one piece no source paper actually implements. This structure — read broadly, then combine and adapt design decisions for a data situation none of the 8 papers directly faced (a Kaggle code competition with multilingual, multi-site, mostly report-only weakly-labeled knee MRI) — is the intended relationship between the source literature and the project, rather than reproducing any one paper's system.

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